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University of Essex | Online

End of Module Assignment: e-Portfolio Submission

Module: Machine Learning
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E-Portfolio Link
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1. OVERVIEW

This e-portfolio presents evidence of my independent work across the Machine Learning module. It highlights my contributions, development of the final project, and structured reflection using the Rolfe model.

The portfolio focuses on building clear, reproducible workflows supported by results, figures, and code. Graphs and tables support the analysis, and all artefacts are linked through the GitHub repository. Writing follows Harvard referencing style and connects learning to real-world applications.

2. REFLECTIVE

1.1 WHAT

During this module, I worked on a machine learning project focused on building and evaluating predictive models. The work involved data preprocessing, feature selection, model training, and performance evaluation. Initially, in Unit 6, the approach included a wide range of features and multiple model types without strong control over validation. This created instability in results and made comparisons difficult.

As part of the team, I contributed to building the regression baseline and participated in evaluating model performance using MAE and R^2 metrics. I also supported the creation of visualisations to communicate results. Collaboration involved sharing updates, reviewing outputs, and aligning tasks across the group.

In Unit 11, the project evolved significantly. The feature set was reduced, noisy variables were removed, and validation techniques were improved by introducing fixed seeds and consistent train-validation-test splits. This resulted in a more stable and reproducible pipeline. My role included refining the model workflow, improving clarity in reporting, and linking results directly to conclusions.

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Beyond technical work, the module exposed me to legal, social, and ethical considerations in machine learning. Issues such as data transparency, bias, and reproducibility became more relevant when working with real-world datasets.

1.2 SO WHAT

This experience significantly improved my understanding of machine learning beyond theory. I developed practical skills in selecting appropriate models, validating results, and interpreting outputs. One key learning point was the importance of starting with a simple baseline model. Previously, I focused on trying complex approaches, but I learned that clear baselines provide better insight into model improvements.

I also improved my ability to handle data correctly. I became more aware of issues such as data leakage, improper splitting, and overfitting. Applying structured validation methods helped me understand how small changes in methodology can significantly impact results.

From a teamwork perspective, I learned the importance of communication and coordination. Initially, overlapping roles caused inefficiencies, but structured updates and clearer responsibilities improved collaboration. Feedback from peers and tutors helped refine both technical work and presentation quality.

Ethically, I developed awareness of the responsibilities involved in machine learning. Model outputs can influence decisions, so ensuring transparency, fairness, and reproducibility is essential. Referencing sources correctly using Harvard style also reinforced academic integrity.

Emotionally, the project was challenging at times, especially when results were inconsistent. However, overcoming these challenges improved my confidence in problem-solving and decision-making.

1.3 NOW WHAT

Going forward, I will apply these lessons in both academic and professional contexts. Technically, I plan to improve model explainability by using techniques such as SHAP

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to better understand feature contributions. I will also explore interpretable models such as Generalised Additive Models (GAMs) to balance performance with clarity.

In terms of validation, I will adopt structured workflows including reproducible experiments, fixed seeds, and clear evaluation metrics. This will ensure reliability in future projects.

For teamwork, I will continue using structured communication methods such as task tracking and regular updates. This will help avoid duplication and improve efficiency.

Professionally, I aim to apply machine learning in real-world engineering contexts, linking my background in electrical systems with data-driven decision-making. The skills developed in this module, including analytical thinking, validation discipline, and structured reporting, are directly transferable to industry roles.

Overall, this module has shifted my approach from simply building models to developing reliable, interpretable, and well-documented solutions.

1.4 Professional Skills Matrix

Skill Area	Baseline	Evidence	Improvement	Next Action
Regression	3	Report & plots	Clear metrics	Apply SHAP
Validation	3	Final runs	Better splits	Checklist
Visualisation	2	Figures	Clear plots	Peer review
Collaboration	4	Notes	Better tracking	Shared tools
Writing	3	Feedback	Clear structure	Harvard checks

3. EVIDENCE OF LEARNING AND CONTRIBUTION

I implemented a structured modelling pipeline with:
- proper train/validation/test splits
- controlled random seeds
- baseline comparison

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Metrics used:
- MAE
- R^2

Figures and tables supported the analysis, ensuring clarity and reproducibility.

Unit 6 Contribution
I built the regression baseline and compared it with a tree model. Metrics were aligned with the problem, and results were visualised clearly.

Figure 1. MAE comparison between models (Linear, Ridge, Random Forest).

Unit 11 Contribution
I refined the pipeline by:
- improving validation
- removing noisy features
- aligning seeds across runs

Figure 2. Feature importance from tree model.

Figure 3. Training vs validation learning curve.

4. EVALUATION: FINAL PROJECT VS ORIGINAL PROPOSAL

The final project improved stability by:
- reducing feature complexity
- improving validation techniques
- focusing on clear metrics

Trade-offs:
- slightly lower accuracy
- higher reliability and interpretability

The pipeline became:
- easier to maintain

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- more reproducible
- clearer for reviewers

Problem – Approach – Impact – Risk – Next Steps

Problem:
Initial approach had too many features and unstable results.

Approach:
- simplify inputs
- standardise preprocessing
- use fixed seeds and fair comparisons

Impact:
- improved stability
- clearer results
- faster review

Risk:
- small drop in accuracy
- limited feature richness

Next Steps:
- SHAP explainability
- GAM comparison
- improved drift checks

5. PRFAQ (PRODUCT REVIEW Q&A)

Q: What changed from the proposal?
A: Reduced features and simplified models to improve stability.

Q: Why this model choice?
A: Linear for clarity, tree for nonlinear patterns.

Q: How were comparisons made fair?

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A: Same seeds, splits, and repeated runs.

Q: Which metrics matter?
A: MAE and R^2 for interpretability.

Q: What improved?
A: Stability, clarity, and reproducibility.

Q: What are limitations?
A: Limited features and need for explainability tools.

Q: How to improve?
A: Use SHAP and GAM models.

6. Summary

The GitHub e-portfolio includes:

- Artefacts from each unit demonstrating learning progress
- Discussion posts and contributions on ethical and professional issues
- Evidence of team collaboration including shared outputs
- Unit 6 (proposal) and Unit 11 (final project) work
- Code, results, and visualisations supporting analysis

All materials are organised to provide clear evidence of development and learning throughout the module.

7. REFERENCES

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